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YOLOV9 Driven Internet of Things Enabled Sustainable Solution for Intelligent Traffic Light Management System for Emergency Vehicles in Large Scale Urban Traffic

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***Abstract*— Due to densely populated urban environment leads to huge traffic in peak hours, Intelligent traffic light management system becomes paramount for emergency vehicle transportation on leveraging the sensor technologies. However sensor data acquired from densely populated urban environment helps to process the traffic congestion based traffic density. Many researches has been carried out to enable intelligent transportation system using internet of things, artificial intelligence and communication technologies but still it requires sustainable solutions for intelligent transportation, traffic congestion management, traffic light controlling with respect to the detection of emergency vehicles like ambulance as it saves the life of the human being. In this paper, AI driven Intelligent of Things enabled sustainable solutions for intelligent traffic light management system for emergency vehicles in the large scale urban traffic. Initially sensor or camera deployed in the smart cities monitors the roads and its surroundings environments. Those acquired information is transmitted to the base station containing IoT servers. In IoT Server, video data is transformed into image frames and processed using YoloV9 based AI model. YoloV9 Model uses multiple component like backbone, neck and head for processing the image frame to recognize and tack the objects in each frame. Especially Backbone model employs convolution neural network for multi scale feature extraction and feature map generation on inclusion of the Generalized Efficient aggregation Network while neck component uses the path aggregation network for future fusion process and head component uses anchor box bounding box prediction method to detect and recognize the object of interest. On detect of the object of interest, distance and speed of the object is computed using gradient flow. Further model incorporates prediction approaches to detected emergency vehicle to estimate its speed and distance from traffic signal as it helps to control traffic light in signal automatically. Furthermore proposed model eliminates the unwanted traffic congestion caused of ambulance vehicle through prioritizing normal traffic in the urban areas. Experimental analysis of the proposed architecture is carried out as IoT prototype to control the traffic signal for emergency vehicles with YoloV9 based AI model using python programming and OpenCV library files. Performance analysis of the proposed model is carried out with respect to object detection accuracy, throughput and latency. Proposed model provides excellent results compared to state of art approaches.**

Keywords: Intelligent Transportation System, YOLOv9 Architecture , Traffic Light Control System , Sustainable Solution, Internet of Things, Artificial intelligence, Emergency Vehicles

I. INTRODUCTION

Densely populated urban cities faces traffic issues due to road congestion in peak hours and it leads to long waiting for emergency vehicles such as ambulance, fire trucks, police. In order to mitigate those challenges many researches has been carried out to establish intelligent transportation system on using monitoring and controlling approaches through incorporation of sensor,

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communication and artificial intelligences technologies . Despite of numerous benefits, it still face several outbreaks in the traffic managements as sensor data acquired from densely populated urban environment helps to process the traffic congestion based traffic density[1].

Hence, sustainable solutions play significant role in large scale urban environment on establishing intelligent transportation, traffic congestion management, traffic light controlling with respect to the detection of emergency vehicles like ambulance as it saves the life of the human being. In this paper, AI driven Intelligent of Things enabled sustainable solutions for intelligent traffic light management system for emergency vehicles in the large scale urban traffic. Initially sensor or camera deployed in the smart cities monitors the roads and its surroundings environments. Those acquired information is transmitted to the base station containing IoT servers. In IoT Server, video data is transformed into image frames and processed using YoloV9 based AI model[2].

YoloV9 Model uses multiple component like backbone, neck and head for processing the image frame to recognize and tack the objects in each frame. Especially Backbone model employs convolution neural network for multi scale feature extraction and feature map generation on inclusion of the Generalized Efficient aggregation Network while neck component uses the path aggregation network for future fusion process and head component uses anchor box bounding box prediction method to detect and recognize the object of interest. On detection of the object of interest, distance and speed of the object is computed using gradient flow. Further model incorporates prediction approaches to detected emergency vehicle to estimate its speed and distance from traffic signal as it helps to control traffic light in signal automatically. Furthermore proposed model eliminates the unwanted traffic congestion caused of ambulance vehicle through prioritizing normal traffic in the urban areas[3].

The remaining article is sectioned as follows, section 2 describes the review of literature related to IoT enabled intelligent transportation system along with object recognition for object detection and object tracking in the densely populated urban regions. Section 3 of the article mentions design architecture of the YoloV9 Driven Internet of Things enabled Sustainable Solution to Intelligent Transportation system towards traffic light controlling for emergency vehicles in the large scale urban environment. Section 4 analyses the experimental results and performance results of the proposed IoT prototype with YoloV9 AI model against the state of art prototypes on basis of object detection accuracy, throughput and latency. Finally section 5 concludes the paper.

II. Related Works

In this section, numerous literature related to Internet of Things enabled intelligent transportation approaches towards traffic light control for emergency vehicles using machine learning approaches are analyzed on multiple aspects were as follows

A. IoT enabled Traffic Light Control System using Convolution N eural Network

In this architecture, IoT prototype is established on incorporating the convolution neural network approach for traffic light control system on identification of the emergency vehicle. IoT Module utilizes the wireless sensor to acquire the data from region of interest and transmit the acquired information to IoT server which is incorporated with Convolution Neural Network for data processing. Convolution Neural Network composed of multiple layer process the data to extract the significant features related to vehicle and establishes the feature map in convolution layer. Further feature map is processed in the fully connected layer using softmax function to detect and categorize the vehicles. On detected emergency vehicle, IoT server transmit the signal to controller of the traffic signal to manage the traffic in the path of the emergency vehicles[4].

B. IoT based Intelligent Traffic Light Control System using Deep Neural Network

In this architecture, IoT prototype is established on incorporating the Deep neural network approach for traffic light control system on identification of the emergency vehicle. IoT Module utilizes the wireless sensor to acquire the data from region of interest and transmit the acquired information to IoT server which is incorporated with Deep Neural Network for data processing. Deep Neural Network composed of multiple layer process the data to extract the significant features related to vehicle and establishes the feature map in the hidden layers. Further feature map is processed in the using softmax function to detect and categorize the vehicles. On detected emergency vehicle, IoT server transmit the signal to controller of the traffic signal to manage the traffic in the path of the emergency vehicles[5].

III. Methodology

A. IoT prototype

In this section, design architecture of the YoloV9 Driven Internet of Things enabled Sustainable Solution to Intelligent Transportation system towards traffic light controlling for emergency vehicles in the large scale urban environment is carried out as follows

IoT Prototype of the proposed architecture composed of the several component such as sensor (camera) modules, controller component, Communication component (transmitter and Receiver) with communication protocol, Traffic lights and web server. Each component specification is described to the intelligent transportation system is as follows

B. Wireless Sensor Module

Wireless sensor is used to monitor and acquire the region of the interest in the IoT enabled intelligent transportation system. In this work, multiple sensor (camera) is deployed in the urban environment to monitor the traffic environment and acquires those environment information in form of digital signal will be transmitted to

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the web server through communication component and its protocol after fusion of the data of different sensor in the controller unit.

C. *Microcontroller Module*

Microcontroller module is used to control to the other component of the IoT prototype. In this work, it carries out hybrid function. Initial function is to gather the monitored information of the sensor and aggregates those information using data aggregation technique before transmitting to base station. Secondary function is to gather the information from data processing unit to control the traffic lights of the specified regions.

D. *Communication Module and Protocol*

Communication Module is used to transmit and receive the data between sensor module and IoT servers. Communication module uses wifi module to carry out full duplex operation towards transmission and reception. Transmission duplex transmit the aggregated data from the controller to base station through effective routing and secure mechanism on the channels of the wifi network. Reception duplex receives the control data from the base station to manage traffic lights.

E. *Web server*

Web server is used to collect and process the data acquired from the sensor field through communication protocol to carry out desired action. It incorporates the Artificial intelligence model for data processing which will detailed sub section of the article. AI model process the collected data and processes the data in form of feature and finally detect the features category to control the traffic light of the specified detected signal.

F. *Traffic Lights*

Traffic lights composed of the different light indicators to control the traffic congestion on reception of the response from the data processing unit. Light Indicator manage the traffic of the specified region on basis of the controller inputs. Figure1 represents the architecture of the IoT enabled traffic management system for emergency vehicles.

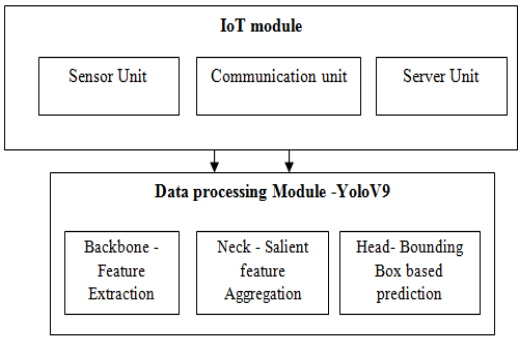


Fig.1. Architecture of the IoT Enabled Sustainable Solution for Intelligent Transportation System

G. *YoloV9 Architecture*

YoloV9 architecture process the data acquired in the base station through its functional components to recognize the object with its moving speed and moving distance based on the spatial coordinates. Yolo9 composed of backbone component, neck component and head component to carry out object recognition task[6] is as follows

H. *Backbone Module*

Backbone Module employs convolution neural network for multi scale feature extraction and feature map generation on inclusion of the Generalized Efficient aggregation Network. Generalized Efficient Aggregation Network composed of multiple computational blocks such as CSP Blocks, Resblocks and Dark Blocks with increasing the computational complexities.

- CSP block : It is efficient in effective gradient flow planning to enhance the feature extraction
- Resblock : It is efficient in image segmentation and feature extraction
- Dark Blocks: It is efficient in generating the feature map to extracted feature in the single frame.

Convolution Layer

YoloV9 backbone utilizes the convolution layer of the cross stage partial Net to downsample the image on minimizing its dimension. Cross stage partial network is computationally efficient as it contains smaller convolutions and smaller kernel size. Kernel size determines the processing speed of the model. Equation 1 extracts the multiscale feature and generate the feature map.

Multiscale Feature map of the image $MF_m(x_i) = K(E(x_i)\text{-}i,j+S)^D$

(1)

Where K is kernel or filter , S is stride and Ex_i is feature

Pooling layers

Pooling Layer is to extract the salient feature of the image frame using Spatial Pyramid Pooling Fast Block. Spatial Pyramid Pooling block uses the attention mechanism to extract the salient feature of the feature map effectively. Salient feature of the image frame is extracted using equation 2.

Salient Feature Map $SF_M = \text{argmax}(MF_m(x_i)$

(2)

I. *Neck module*

Neck component uses the path aggregation network for feature fusion process to the feature extracted in different image scales. Path Aggregation Network[7] process the feature in the pyramid structure and spatial structure for feature aggregation as it is more faster in feature aggregation process. Aggregated features and its feature map is represented in the equation 3

Aggregated feature $AF_m = \text{Argmax}(P_s(SF_M)+ S_s(SF_M))$

(3)

- Pyramid Structure

Pyramid structure gathers the features in top down approach to preserve the semantic information and gathers features in bottom up approach to preserve the lateral information's.

- Spatial Structure

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Spatial structure gathers the fine grained and coarse grained features of the multiscale feature through kernel and stride.

J. *Head Module*

Head Module uses anchor box bounding box based prediction method[8] to detect and recognize the object of interest on the concatenated feature map. On detection of the object of interest, distance and speed of the object is computed using gradient flow. Gradient flow is employed to calculate the larger correlation variation among the objects in the image.

- Sigmoid Linear Unit Block

Head includes normalization and activation blocks to refine the concatenated feature maps of the salient features towards object detection on utilizing the Sigmoid Linear Unit[9] activation function for non – linearity of the salient features management.

K. *Prediction Mechanism - Fully Connected layer*

Fully connected layer of Yolov9 backbone model incorporates uses the soft max function to the prediction task. Prediction task detects the objects considered as emergency vehicle along its estimate in terms of the its speed and distance from traffic signal. Those computation helps to control traffic light in signal automatically through transmission of the control signal to the controller. Furthermore proposed model is highly capable in eliminating the unwanted traffic congestion caused of ambulance vehicle through prioritizing normal traffic in the urban areas. Softmax function is employed using support vector machine.

Classification Function $C = \{(x,y)|x_i \varepsilon R^N \; y_1 \varepsilon \{1,-1\}^N$
Where R is the Decision Boundary and N support vector of the different hyperplane.

Algorithm 1: YoloV9 based Object Detection
Input: Video data of the Region of Interest
Output : Object Detection with Its moving speed and Distance from Nearest signal
1:Process()
2:Video Conversion()
Video Sequence into Image frames V_s ={Frame 1, Frame 2, Frame 3...Frame N}
3:Image Frame Augmentation ()
4:Rotate(Frames)
5:YoloV9()
6:Backbone_ Convolutional Neural Network()
7:CSP_Block_Feature Extraction()
8:Transformation Image into gradient flow
Convolution layer (Kernel, Stride, Image)
9:Multiscale feature map
Spatial Pyramid Pooling Fast Block ()
10:Salient feature Map
11:Neck(Salient feature Map)
12:Path Aggregation Network ()
13:Feature Aggregation = Pyramid structure (salient feature) + Spatial Structure(Salient feature)
14:Head(Aggregated features)
15:Bounding box (Aggregated features)
16:Batch Normalization (Sigmoid Linear Unit)
17:Fully Connected Layer()
18:Softmax function (support Vector machine)
19:Hyperplane{Normalized Features)
20:Classification Function (Decision Boundaries)
21:Object Class = {Signal , Normal Vehicles, Emergency Vehicles)
22:Traffic Light Control Strategies()
Fix threshold for distance as T_D and Speed as S_D
23:if (object = = Emergency Vehicle)
Compute the Distance of the vehicle object from the signal using gradient flow
Compute Speed of the vehicle object using Gradient Flow
Nested
24:If(Distance < Threshold T_D & Speed < Threshold Ts)
Change Signal to control traffic in the path of the emergency vehicles.

Proposed model eliminates the unwanted traffic congestion caused of ambulance vehicle through prioritizing normal traffic in the urban areas

IV. . Experimental Analysis

Experimental analysis of the IoT enabled Sustainable Solution for Intelligent Transportation System for emergency vehicle in large scale urban traffic. In this IoT prototype composed of multiple sensor acquires the traffic information and controls the traffic signal for emergency vehicle such as ambulance on processing the acquired traffic information in the YoloV9 based AI model using python programming and OpenCV library files.[10] Performance analysis of the proposed model is carried out with respect to object detection accuracy, throughput and latency. Proposed model provides excellent results compared to state of art approaches. Table 1 mentions hyperparameter values of the YoloV9 model

TABLE I. Hyperparameter of YOLOV9 Model

Parameter	Value
Learning Rate	10^{-7}
Batch Size	50 images frames
Epoch	200
Momentum	0.788
Weight Decay	0.007

Input video of the coco dataset is transformed into image frames and each image frame is processed using yolov9 model to extract the salient feature with bounding box in order to predicts the speed and distance of the vehicle through decision boundaries of the gradient flow. Especially feature is processed in form the aggregated feature map on spatial and pyramid structures. Figure 2 represents the moving vehicle with speed computation.

Yolov9 model provides excellent performance in emergency vehicle detection a long distance and speed of it to control the traffic signal of large scale urban traffic. Further response time of the model is computed with respect of the throughput and latency as it determines the computation overhead of the model.

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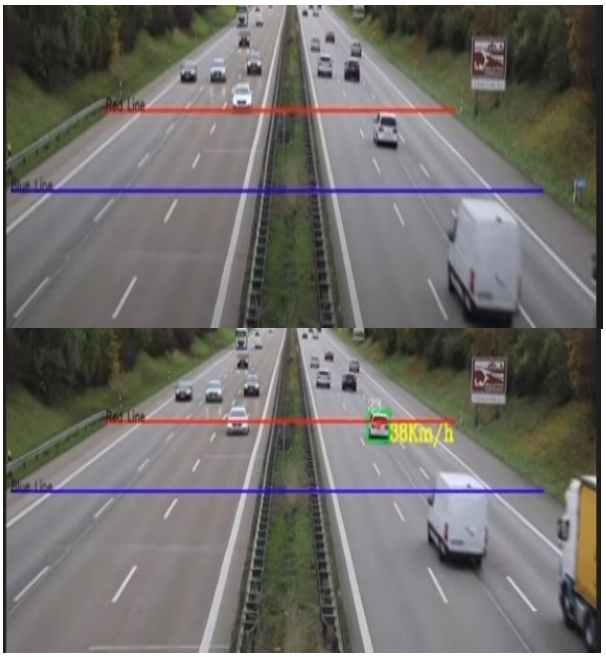


Fig.2. Bounding box based Moving object Recognition with speed.

Table 2 provides the performance evaluation of the current model against state of art approaches with respect to the accuracy, throughput and latency.

TABLE II. Performance Evaluation

Technique	Class	Object Detection Accuracy	Throughput	Latency
IoT Enabled YoloV9-Proposed	Emergency Vehicle	99.8	80kbps	80ms
	Normal Vehicle	98.9		
IoT Enabled CNN-Exiting 1	Emergency Vehicle	94.8	62kbps	90ms
	Normal Vehicle	93.6		
IoT Enabled DNN-Existing 3	Emergency Vehicle	91.8	49ms	120ms
	Normal Vehicle	92.4		

Finally, Yolov9 model for object detection and classification on various categories performance excellent against various state of art approaches on accuracy, throughput and latency were represented in the figure 3.

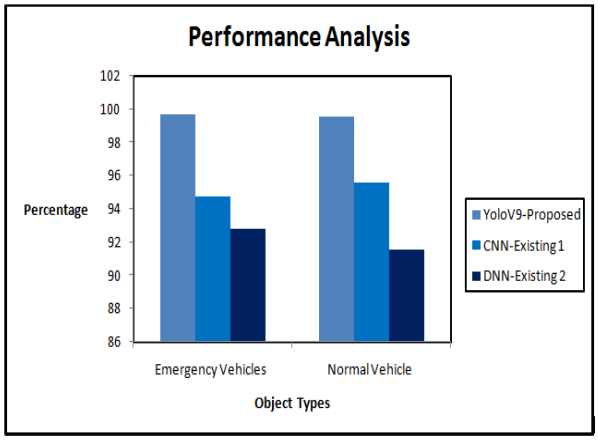


Fig.3. Performance analysis of the model with respect to Accuracy

CONCLUSION

In this article, YoloV9 driven Internet of Things enabled sustainable solutions for intelligent traffic light management system for emergency vehicles in the large scale urban traffic has been designed and developed as IoT prototype with object recognition approaches. IoT module acquires the sensor information and processes those information in server using the yolov9 based AI model. YoloV9 modules use the backbone component with Generalized Efficient Aggregation Network for feature extraction, neck component with path aggregation network for feature extraction and head component with bounding box based prediction method to detect and classify the object. Detected objects were estimated with distance and speed. Finally control strategies uses those estimates to control the traffic light automatically. Furthermore, proposed model eliminates the unwanted traffic congestion caused of ambulance vehicle through prioritizing normal traffic in the urban areas. Experimental analysis and performance analysis of the proposed architecture to control the traffic signal for emergency vehicles provides excellent results compared to state of art approaches.

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